

Lab Assignment 5: Image and Multiclass Classification: Digits and Iris

CSAI2017P: Applied Machine Learning (Lab)

Experiments 6–7 (Unit VI) • CO2, CO4

Instructor: Dr. Tofik Ali

Due: announced in the lab

Student Name: _____

SAP ID: _____

Batch: _____

Lab quiz: LQ5, after submission

Objective

- Treat an image as a feature vector and classify it.
- Read a multiclass confusion matrix and find the confusable pairs.
- Compare a distance-based, a margin-based and a tree-based classifier on the same data.

Datasets used in this assignment

B1 (scikit-learn digits), A-small (Iris).

Full descriptions, sources and loading snippets for all anchor datasets are in **Anchor-Datasets.pdf**, alongside this brief.

Instructions

- Use a train/test split (80/20 unless stated) and set `random_state` for reproducibility.
- Fit every transformer inside a `Pipeline` so nothing is fitted on the test set.
- Report the metric named in the question. A number without units or context earns no marks.
- Every question asks for a short written observation. Code without commentary caps you at half marks.
- The lab quiz that follows will ask you to read, trace and modify *your own* submitted code.

Submit:

- Notebook: `AML_LA05_<SAPID>.ipynb` with all cells executed and outputs visible.
- Report: 2–3 pages PDF, or a `README.md`, carrying your results table and observations.
- Do not upload datasets. Cite the source and include the loading code.

Section	Level	Choice rule	Marks
A	Easy	Attempt any 2 of 3 (2 marks each)	4
B	Medium	Attempt any 1 of 2 (4 marks each)	4
C	Hard	Compulsory	2
Total			10

Course outcomes assessed by this assignment: **CO2, CO4**. Attempting more than the choice rule allows is not penalised; the best attempts are marked.

Section A (Easy) — attempt any 2 of 3 ($2 \times 2 = 4$)**A1. Iris in ten lines** [2 marks]

Dataset: Iris (load_iris).

- (a) Split 75/25 stratified, scale, and fit `KNeighborsClassifier(n_neighbors=5)`.
- (b) Report test accuracy and the confusion matrix.
- (c) Name the two species that get confused and state which measurement separates them best.

A2. Digits as vectors [2 marks]

Dataset: B1 (digits).

- (a) Display any five images with their labels using `imshow`.
- (b) State the shape of one image and the shape of the flattened feature matrix.
- (c) In two lines, state what information the flattening throws away.

A3. Choosing k [2 marks]

Dataset: B1 (digits).

- (a) Fit KNN for $k \in \{1, 3, 5, 7, 9, 11, 15\}$ and plot test accuracy against k .
- (b) Report the best k .
- (c) Explain in two lines what goes wrong at $k = 1$ and what goes wrong at a very large k .

Section B (Medium) — attempt any 1 of 2 ($1 \times 4 = 4$)**B1. Three classifiers, one dataset** [4 marks]

Dataset: B1 (digits).

- (a) Train KNN, an SVM with an RBF kernel and a random forest on the same split.
- (b) Report accuracy and macro-F1 for each, plus fit time.
- (c) Recommend one for a device with tight memory and explain in 4–6 lines.

B2. Which digits get confused [4 marks]

Dataset: B1 (digits).

- (a) Take your best model and plot the full 10×10 confusion matrix as a heatmap.
- (b) Report the three most frequently confused ordered pairs.
- (c) Display four misclassified images and explain in 4–6 lines what makes them hard.

Section C (Hard) — compulsory (2)**C1. Does more resolution help** [2 marks]

Datasets: B1 and B2.

- (a) Train your best model on the 8×8 digits and on a 10,000-row subsample of 28×28 MNIST.
- (b) Report accuracy and fit time for both, and comment in 4–6 lines on what the extra resolution bought.

End of Lab Assignment 5

Sources: Müller and Guido, *Introduction to Machine Learning with Python*, companion notebooks.;McKinney, *Python for Data Analysis* (3e), O'Reilly, 2022.;Scikit-learn user guide, accessed 2026-08-04.;Matplotlib and Seaborn documentation, accessed 2026-08-04.